🧠AI Trust Operations Platform

Architecture Deep Dive — Component by Component + Data Flow

> **Purpose**: This document provides a complete, printable, and shareable architectural explanation of the AI Trust Operations Platform, covering all components, responsibilities, and data flows.

👎🤩High-Level System Architecture

The platform is built as **layered, deterministic, human-governed AI architecture**.

```
External Systems (Orders / Escrow / Claims)
             ↓
TRUST-OPS-010 — Event & Audit Backbone
             ↓
AI-TRUST Layer (Risk, Behavior, Decisions)
             ↓
AI-OPS Governance Layer (Monitoring, Feedback, Change)
             ↓
Human Trust Operations
```

Each layer is **read-only upward**, **auditable**, and **no layer auto-enforces actions**.

🧠🤪AI-TRUST-001 — Deterministic Risk Scoring Engine

😖Purpose

Quantify transactional and seller risk into a **single deterministic score (0–100)**.

👇Inputs

- Order completion rate
- Escrow release/refund history
- Claims & disputes
- Transaction velocity
- Seller account age

🖕Processing Logic

- Weighted scoring rules

- Threshold mapping (LOW / MEDIUM / HIGH)
- No ML, no learning, same input = same output

👇🏽Outputs

```
{
  "riskScore": 74,
  "riskLevel": "HIGH",
  "confidence": 0.89
}
```

👎Architectural Role

> Answers: **"How risky is this entity right now?"**

👉🏿🤫AI-TRUST-002 — Seller Behavior & Pattern Detection

😖Purpose

Detect **behavioral anomalies** not visible in raw scores.

👇🏾Inputs

- TRUST-OPS-010 event stream
- Time-based seller activity
- Order & refund sequences

🖕🏼Processing Logic

- Deterministic pattern rules
- Frequency + severity scoring
- Pattern confidence calculation

👇🏽Outputs

```
{
  "behaviorRiskScore": 0.63,
  "detectedPatterns": ["VELOCITY_SPIKE", "REFUND_LOOP"]
}
```

👎Architectural Role

> Answers: **"Is seller behavior changing in a risky way?"**

🤬TRS-002 — Seller Trust Score System

😖Purpose

Maintain **long-term seller reputation** independent of single transactions.

👇🏾Inputs

- Order success history
- Claim resolutions
- Refund ratios
- Consistency over time

🖕🏼Processing Logic

- Weighted trust formula
- Decay-aware history
- Event-driven updates

👇🏽Outputs

```
{
  "trustScore": 82,
  "trend": "STABLE"
}
```

👎Architectural Role

> Answers: **"Is this seller historically trustworthy?"**

🧠🤭AI-TRUST-003 — Decision & Recommendation Engine

😖Purpose

Synthesize all trust signals into **explainable, advisory-only recommendations**.

👇🏾Inputs

- AI-TRUST-001 risk score
- AI-TRUST-002 behavior risk
- TRS-002 trust score
- TRUST-OPS-010 events
- Order context (value, buyer history)

🖕🏼 Processing Logic

- Weighted synthesis (deterministic)
- Escalation mapping
- Rule-trigger evaluation

👇🏽 Outputs

```
{
  "escalationLevel": "REVIEW",
  "decisionConfidence": 0.86,
  "recommendedActions": ["MANUAL_REVIEW", "ESCROW_ENFORCEMENT"],
  "reasoning": "High risk + declining trust trend"
}
```

👎 Architectural Role

Answers: **"What should humans consider doing?"**

😾🤮 TRUST-OPS-010 — Trust Events & Audit Backbone

😖 Purpose

Serve as the **single source of truth** for trust-related events.

Responsibilities

- Event ingestion
- Event normalization
- Cross-system audit logging

Example Events

- escrow_released
- refund_issued
- claim_opened
- decision_override

👎 Architectural Role

Nervous system of the platform — everything flows through it.

👹🤯AI-OPS-004 — Monitoring, Drift & Explainability

😖Purpose

Ensure AI remains **stable, fair, and explainable over time**.

Capabilities

- Drift detection (scores, patterns, decisions)
- Model health scoring (0–100)
- Historical explainability timelines

Outputs

- Drift reports
- aiHealthScore
- Decision change explanations

👎Architectural Role

> "Is AI still behaving correctly?"

🧠🥰AI-OPS-005 — Human Feedback & Continuous Improvement

😖Purpose

Capture **human judgment** and measure AI-human alignment.

Capabilities

- Feedback capture
- Override analysis
- Trust-in-AI metrics
- Improvement signals (read-only)

👎Architectural Role

> "Do humans trust AI — and where does it fail?"

😉🥱AI-OPS-006 — Controlled Change & Rollback Framework

😖Purpose

Allow **safe AI evolution** without production risk.

Capabilities

- Change proposals
- Shadow evaluation
- Approval workflow
- Instant rollback

👎Architectural Role

> "How do we safely change AI without breaking trust?"

😌1🥲End-to-End Data Flow Example

1. Order event → TRUST-OPS-010
2. Risk scored (001)
3. Behavior analyzed (002)
4. Trust score referenced (TRS)
5. Decision generated (003)
6. Logged + monitored (004)
7. Human feedback captured (005)
8. Future changes governed (006)

👈🏼Final Summary

This architecture creates a **Regulated, Explainable, Human-Governed AI Trust Operating System** where:

- AI never auto-enforces
- Humans remain accountable
- Every decision is explainable
- Every change is reversible

Document Status: ✅Ready for download, sharing, and implementation reference.